Constructions With Relations

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00NE This chapter contains some material about constructions with relations. Notably, we discuss and explore:

029U 1. The existence or non-existence of Kan extensions and Kan lifts in the 2-category **Rel** (Section 9.2).

029V 2. The various kinds of constructions involving relations, such as graphs, domains, ranges, unions, intersections, products, converse relations, composition of relations, and collages (Section 9.3).

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00NF 9.1 Co/Limits in the Category of Relations

This section is currently just a stub, and will be properly developed later on.

00NG 9.2 Kan Extensions and Kan Lifts in the 2-Category of Relations

00NH 9.2.1 Left Kan Extensions in Rel

00NJ **Proposition 9.2.1.1.1.** Let $R\colon A \nrightarrow B$ be a relation.

00NK 1. *Non-Existence of All Left Kan Extensions in* **Rel**. Not all relations in **Rel** admit left Kan extensions.

00NL 2. *Characterisation of Relations Admitting Left Kan Extensions Along Them*. The following conditions are equivalent:

02A1 (a) The left Kan extension

$$\mathrm{Lan}_R\colon \mathbf{Rel}(A, X) \to \mathbf{Rel}(B, X)$$

along R exists.

02A2 (b) The relation R admits a left adjoint in **Rel**.

02A3 (c) The relation R is of the form f^{-1} (as in **??**) for some function f.

Proof. *Item 1*, *Non-Existence of All Left Kan Extensions in* **Rel**: Omitted, but will eventually follow Fosco Loregian's comment on [MO 460656].
Item 2, *Characterisation of Relations Admitting Left Kan Extensions Along Them*: Omitted, but will eventually follow Tim Campion's answer to to [MO 460656]. □

00NM **Question 9.2.1.1.2.** Given relations $S\colon A \nrightarrow X$ and $R\colon A \nrightarrow B$, is there a characterisation of when the left Kan extension

$$\mathrm{Lan}_S(R)\colon B \nrightarrow X$$

exists in terms of properties of R and S?

This question also appears as [MO 461592].

00NN **Question 9.2.1.1.3.** As shown in Item 2 of Definition 9.2.1.1.1, the left Kan extension

$$\mathrm{Lan}_R \colon \mathbf{Rel}(A, X) \to \mathbf{Rel}(B, X)$$

along a relation of the form $R = f^{-1}$ exists. Is there a explicit description of it, similarly to the explicit description of right Kan extensions given in Definition 9.2.3.1.1?

This question also appears as [MO 461592].

00NP 9.2.2 Left Kan Lifts in Rel

00NQ **Proposition 9.2.2.1.1.** Let $R \colon A \nrightarrow B$ be a relation.

00NR 1. *Non-Existence of All Left Kan Lifts in* **Rel**. Not all relations in **Rel** admit left Kan lifts.

00NS 2. *Characterisation of Relations Admitting Left Kan Lifts Along Them*. The following conditions are equivalent:

02A4 (a) The left Kan lift

$$\mathrm{Lift}_R \colon \mathbf{Rel}(X, B) \to \mathbf{Rel}(X, A)$$

along R exists.

02A5 (b) The relation R admits a right adjoint in **Rel**.

02A6 (c) The relation R is of the form $\mathrm{Gr}(f)$ (as in **??**) for some function f.

Proof. *Item 1*, *Non-Existence of All Left Kan Lifts in* **Rel**: Omitted, but will eventually follow (the dual of) Fosco Loregian's comment on [MO 460656].
Item 2, *Characterisation of Relations Admitting Left Kan Lifts Along Them*: Omitted, but will eventually follow Tim Campion's answer to to [MO 460656]. □

00NT **Question 9.2.2.1.2.** Given relations $S \colon A \nrightarrow X$ and $R \colon A \nrightarrow B$, is there a characterisation of when the left Kan lift

$$\mathrm{Lift}_S(R) \colon X \nrightarrow A$$

exists in terms of properties of R and S?

This question also appears as [MO 461592].

00NU **Question 9.2.2.1.3.** As shown in Item 2 of Definition 9.2.2.1.1, the left Kan lift

$$\mathrm{Lift}_R\colon \mathbf{Rel}(X,B) \to \mathbf{Rel}(X,A)$$

along a relation of the form $R = \mathrm{Gr}(f)$ exists. Is there a explicit description of it, similarly to the explicit description of right Kan lifts given in Definition 9.2.4.1.1?

This question also appears as [MO 461592].

00NV 9.2.3 Right Kan Extensions in Rel

Let $R\colon A \nrightarrow B$ be a relation.

00NW **Proposition 9.2.3.1.1.** The right Kan extension

$$\mathrm{Ran}_R\colon \mathrm{Rel}(A,X) \to \mathrm{Rel}(B,X)$$

along R in **Rel** exists and is given by

$$\mathrm{Ran}_R(S) \overset{\text{def}}{=} \int_{a\in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^{-2}, S_a^{-1}\right)$$

for each $S \in \mathrm{Rel}(A,X)$, so that the following conditions are equivalent:

02A7 1. We have $b \sim_{\mathrm{Ran}_R(S)} x$.

02A8 2. For each $a \in A$, if $a \sim_R b$, then $a \sim_S x$.

Proof. We have

$$\begin{aligned}
\mathrm{Hom}_{\mathbf{Rel}(A,X)}(S \diamond R, T) &\cong \int_{a\in A}\int_{x\in X} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left((S\diamond R)_a^x, T_a^x\right)\\
&\cong \int_{a\in A}\int_{x\in X} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(\left(\int^{b\in B} S_b^x \times R_a^b\right), T_a^x\right)\\
&\cong \int_{a\in A}\int_{x\in X}\int_{b\in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S_b^x \times R_a^b, T_a^x\right)\\
&\cong \int_{a\in A}\int_{x\in X}\int_{b\in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S_b^x, \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^b, T_a^x\right)\right)\\
&\cong \int_{b\in B}\int_{x\in X}\int_{a\in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S_b^x, \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^b, T_a^x\right)\right)\\
&\cong \int_{b\in B}\int_{x\in X} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S_b^x, \int_{a\in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^b, T_a^x\right)\right)
\end{aligned}$$

$$\cong \mathrm{Hom}_{\mathbf{Rel}(B,X)}\left(S, \int_{a\in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^{-2}, T_a^{-1}\right)\right)$$

naturally in each $S \in \mathbf{Rel}(B, X)$ and each $T \in \mathbf{Rel}(A, X)$, showing that

$$\int_{a\in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^{-2}, T_a^{-1}\right)$$

is right adjoint to the precomposition functor $- \diamond R$, being thus the right Kan extension along R. Here we have used the following results, respectively (i.e. for each $\cong$ sign):

02A9 1. Relations, Item 1 of Definition 8.1.1.1.8.

02AA 2. **??**.

02AB 3. Ends and Coends, **??** of **??**.

02AC 4. Sets, Definition 3.2.2.1.5.

02AD 5. Ends and Coends, **??** of **??**.

02AE 6. Ends and Coends, **??** of **??**.

02AF 7. Relations, Item 1 of Definition 8.1.1.1.8.

This finishes the proof. □

00NX 9.2.4 Right Kan Lifts in Rel

Let $R\colon A \nrightarrow B$ be a relation.

00NY **Proposition 9.2.4.1.1.** The right Kan lift

$$\mathrm{Rift}_R\colon \mathrm{Rel}(X, B) \longrightarrow \mathrm{Rel}(X, A)$$

along R in **Rel** exists and is given by

$$\mathrm{Rift}_R(S) \overset{\mathrm{def}}{=} \int_{b\in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_{-_1}^b, S_{-_2}^b\right)$$

for each $S \in \mathrm{Rel}(X, B)$, so that the following conditions are equivalent:

02AG 1. We have $x \sim_{\mathrm{Rift}_R(S)} a$.

02AH 2. For each $b \in B$, if $a \sim_R b$, then $x \sim_S b$.

Proof. We have

$$\begin{aligned}
\mathrm{Hom}_{\mathbf{Rel}(X,B)}(R \diamond S, T) &\cong \int_{x \in X} \int_{b \in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left((R \diamond S)^b_x, T^b_x\right) \\
&\cong \int_{x \in X} \int_{b \in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(\left(\int^{a \in A} R^b_a \times S^a_x\right), T^b_x\right) \\
&\cong \int_{x \in X} \int_{b \in B} \int_{a \in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_a \times S^a_x, T^b_x\right) \\
&\cong \int_{x \in X} \int_{b \in B} \int_{a \in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S^a_x, \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_a, T^b_x\right)\right) \\
&\cong \int_{x \in X} \int_{a \in A} \int_{b \in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S^a_x, \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_a, T^b_x\right)\right) \\
&\cong \int_{x \in X} \int_{a \in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(S^a_x, \int_{b \in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_a, T^b_x\right)\right) \\
&\cong \mathrm{Hom}_{\mathbf{Rel}(X,A)}\left(S, \int_{b \in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_{-_1}, T^b_{-_2}\right)\right)
\end{aligned}$$

naturally in each $S \in \mathbf{Rel}(X, A)$ and each $T \in \mathbf{Rel}(X, B)$, showing that

$$\int_{b \in B} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R^b_{-_1}, S^b_{-_2}\right)$$

is right adjoint to the postcomposition functor $R \diamond -$, being thus the right Kan lift along R. Here we have used the following results, respectively (i.e. for each $\cong$ sign):

02AJ 1. Relations, Item 1 of Definition 8.1.1.1.8.

02AK 2. **??**.

02AL 3. Ends and Coends, **??** of **??**.

02AM 4. Sets, Definition 3.2.2.1.5.

02AN 5. Ends and Coends, **??** of **??**.

02AP 6. Ends and Coends, **??** of **??**.

02AQ 7. Relations, Item 1 of Definition 8.1.1.1.8.

This finishes the proof. □

00NZ 9.3 More Constructions With Relations

00PM 9.3.1 The Domain and Range of a Relation

Let A and B be sets.

00PN **Definition 9.3.1.1.1.** Let $R \subset A \times B$ be a relation.[1,2]

02AV 1. The **domain of** R is the subset $\mathrm{dom}(R)$ of A defined by

$$\mathrm{dom}(R) \overset{\text{def}}{=} \left\{ a \in A \,\middle|\, \begin{array}{l} \text{there exists some } b \in B \\ \text{such that } a \sim_R b \end{array} \right\}.$$

02AW 2. The **range of** R is the subset $\mathrm{range}(R)$ of B defined by

$$\mathrm{range}(R) \overset{\text{def}}{=} \left\{ b \in B \,\middle|\, \begin{array}{l} \text{there exists some } a \in A \\ \text{such that } a \sim_R b \end{array} \right\}.$$

[1] Following **??**, **??**, we may compute the (characteristic functions associated to the) domain and range of a relation using the following colimit formulas:

$$\begin{aligned} \chi_{\mathrm{dom}(R)}(a) &\cong \operatorname*{colim}_{b \in B}\left(R_a^b\right) && (a \in A) \\ &\cong \bigvee_{b \in B} R_a^b, \\ \chi_{\mathrm{range}(R)}(b) &\cong \operatorname*{colim}_{a \in A}\left(R_a^b\right) && (b \in B) \\ &\cong \bigvee_{a \in A} R_a^b, \end{aligned}$$

where the join $\bigvee$ is taken in the poset $(\{\mathrm{true}, \mathrm{false}\}, \preceq)$ of Constructions With Sets, Definition 3.2.2.1.3.

[2] Viewing R as a function $R\colon A \to \mathcal{P}(B)$, we have

$$\begin{aligned} \mathrm{dom}(R) &\cong \operatorname*{colim}_{y \in Y}(R(y)) \\ &\cong \bigcup_{y \in Y} R(y), \\ \mathrm{range}(R) &\cong \operatorname*{colim}_{x \in X}(R(x)) \\ &\cong \bigcup_{x \in X} R(x), \end{aligned}$$

00PP 9.3.2 Binary Unions of Relations

Let A and B be sets and let R and S be relations from A to B.

00PQ **Definition 9.3.2.1.1.** The **union of R and S**[3] is the relation $R \cup S$ from A to B defined as follows:

- Viewing relations from A to B as subsets of $A \times B$, we define[4]

$$R \cup S \overset{\text{def}}{=} \{(a, b) \in B \times A \mid \text{we have } a \sim_R b \text{ or } a \sim_S b\}.$$

- Viewing relations from A to B as functions $A \longrightarrow \mathcal{P}(B)$, we define

$$[R \cup S](a) \overset{\text{def}}{=} R(a) \cup S(a)$$

  for each $a \in A$.

00PR **Proposition 9.3.2.1.2.** Let R, S, R_1, and R_2 be relations from A to B, and let S_1 and S_2 be relations from B to C.

00PS 1. *Interaction With Converses.* We have

$$(R \cup S)^\dagger = R^\dagger \cup S^\dagger.$$

00PT 2. *Interaction With Composition.* We have

$$(S_1 \diamond R_1) \cup (S_2 \diamond R_2) \overset{\text{poss.}}{\neq} (S_1 \cup S_2) \diamond (R_1 \cup R_2).$$

Proof. *Item 1*, *Interaction With Converses*: Clear.
Item 2, *Interaction With Composition*: Unwinding the definitions, we see that:

- The condition for $(S_1 \diamond R_1) \cup (S_2 \diamond R_2)$ is:
  - There exists some $b \in B$ such that:
    - $a \sim_{R_1} b$ and $b \sim_{S_1} c$;

    or

    - $a \sim_{R_2} b$ and $b \sim_{S_2} c$;

[3] *Further Terminology:* Also called the **binary union of R and S**, for emphasis.
[4] This is the same as the union of R and S as subsets of $A \times B$.

- The condition for $(S_1 \cup S_2) \diamond (R_1 \cup R_2)$ is:
  - There exists some $b \in B$ such that:
    * $a \sim_{R_1} b$ or $a \sim_{R_2} b$;

    and
    * $b \sim_{S_1} c$ or $b \sim_{S_2} c$.

These two conditions may fail to agree (counterexample omitted), and thus the two resulting relations on $A \times C$ may differ. □

00PU 9.3.3 Unions of Families of Relations

Let A and B be sets and let $\{R_i\}_{i \in I}$ be a family of relations from A to B.

00PV **Definition 9.3.3.1.1.** The **union of the family** $\{R_i\}_{i \in I}$ is the relation $\bigcup_{i \in I} R_i$ from A to B defined as follows:

- Viewing relations from A to B as subsets of $A \times B$, we define[5]

$$\bigcup_{i \in I} R_i \stackrel{\text{def}}{=} \left\{ (a, b) \in (A \times B)^{\times I} \;\middle|\; \begin{array}{l} \text{there exists some } i \in I \\ \text{such that } a \sim_{R_i} b \end{array} \right\}.$$

- Viewing relations from A to B as functions $A \to \mathcal{P}(B)$, we define

$$\left[\bigcup_{i \in I} R_i \right](a) \stackrel{\text{def}}{=} \bigcup_{i \in I} R_i(a)$$

  for each $a \in A$.

00PW **Proposition 9.3.3.1.2.** Let A and B be sets and let $\{R_i\}_{i \in I}$ be a family of relations from A to B.

00PX 1. *Interaction With Converses.* We have

$$\left(\bigcup_{i \in I} R_i \right)^{\dagger} = \bigcup_{i \in I} R_i^{\dagger}.$$

Proof. *Item 1*, *Interaction With Converses*: Clear. □

[5] This is the same as the union of $\{R_i\}_{i \in I}$ as a collection of subsets of $A \times B$.

00PY 9.3.4 Binary Intersections of Relations

Let A and B be sets and let R and S be relations from A to B.

00PZ **Definition 9.3.4.1.1.** The **intersection of R and S**[6] is the relation $R \cap S$ from A to B defined as follows:

- Viewing relations from A to B as subsets of $A \times B$, we define[7]

$$R \cap S \stackrel{\text{def}}{=} \{(a, b) \in B \times A \mid \text{we have } a \sim_R b \text{ and } a \sim_S b\}.$$

- Viewing relations from A to B as functions $A \longrightarrow \mathcal{P}(B)$, we define

$$[R \cap S](a) \stackrel{\text{def}}{=} R(a) \cap S(a)$$

  for each $a \in A$.

00Q0 **Proposition 9.3.4.1.2.** Let R, S, R_1, and R_2 be relations from A to B, and let S_1 and S_2 be relations from B to C.

00Q1 1. *Interaction With Converses.* We have

$$(R \cap S)^\dagger = R^\dagger \cap S^\dagger.$$

00Q2 2. *Interaction With Composition.* We have

$$(S_1 \diamond R_1) \cap (S_2 \diamond R_2) = (S_1 \cap S_2) \diamond (R_1 \cap R_2).$$

Proof. *Item 1*, *Interaction With Converses*: Clear.
Item 2, *Interaction With Composition*: Unwinding the definitions, we see that:

- The condition for $(S_1 \diamond R_1) \cap (S_2 \diamond R_2)$ is:
  - There exists some $b \in B$ such that:
    - $a \sim_{R_1} b$ and $b \sim_{S_1} c$;

    and
    - $a \sim_{R_2} b$ and $b \sim_{S_2} c$;

[6] *Further Terminology:* Also called the **binary intersection of R and S**, for emphasis.
[7] This is the same as the intersection of R and S as subsets of $A \times B$.

- The condition for $(S_1 \cap S_2) \diamond (R_1 \cap R_2)$ is:
  - There exists some $b \in B$ such that:
    - $a \sim_{R_1} b$ and $a \sim_{R_2} b$;

    and
    - $b \sim_{S_1} c$ and $b \sim_{S_2} c$.

These two conditions agree, and thus so do the two resulting relations on $A \times C$. □

00Q3 9.3.5 Intersections of Families of Relations

Let A and B be sets and let $\{R_i\}_{i \in I}$ be a family of relations from A to B.

00Q4 **Definition 9.3.5.1.1.** The **intersection of the family** $\{R_i\}_{i \in I}$ is the relation $\bigcup_{i \in I} R_i$ defined as follows:

- Viewing relations from A to B as subsets of $A \times B$, we define[8]

$$\bigcup_{i \in I} R_i \overset{\text{def}}{=} \left\{ (a, b) \in (A \times B)^{\times I} \;\middle|\; \begin{array}{l} \text{for each } i \in I, \\ \text{we have } a \sim_{R_i} b \end{array} \right\}.$$

- Viewing relations from A to B as functions $A \longrightarrow \mathcal{P}(B)$, we define

$$\left[\bigcap_{i \in I} R_i \right](a) \overset{\text{def}}{=} \bigcap_{i \in I} R_i(a)$$

  for each $a \in A$.

00Q5 **Proposition 9.3.5.1.2.** Let A and B be sets and let $\{R_i\}_{i \in I}$ be a family of relations from A to B.

00Q6 1. *Interaction With Converses.* We have

$$\left(\bigcap_{i \in I} R_i \right)^{\dagger} = \bigcap_{i \in I} R_i^{\dagger}.$$

Proof. *Item 1*, *Interaction With Converses*: Clear. □

[8] This is the same as the intersection of $\{R_i\}_{i \in I}$ as a collection of subsets of $A \times B$.

00Q7 9.3.6 Binary Products of Relations

Let A, B, X, and Y be sets, let $R\colon A \nrightarrow B$ be a relation from A to B, and let $S\colon X \nrightarrow Y$ be a relation from X to Y.

00Q8 **Definition 9.3.6.1.1.** The **product of R and S**[9] is the relation $R \times S$ from $A \times X$ to $B \times Y$ defined as follows:

- Viewing relations from $A \times X$ to $B \times Y$ as subsets of $(A \times X) \times (B \times Y)$, we define $R \times S$ as the Cartesian product of R and S as subsets of $A \times X$ and $B \times Y$.[10]

- Viewing relations from $A \times X$ to $B \times Y$ as functions $A \times X \to \mathcal{P}(B \times Y)$, we define $R \times S$ as the composition

$$A \times X \xrightarrow{R \times S} \mathcal{P}(B) \times \mathcal{P}(Y) \overset{\mathcal{P}^{\otimes}_{B,Y}}{\hookrightarrow} \mathcal{P}(B \times Y)$$

  in Sets, i.e. by

$$[R \times S](a, x) \overset{\text{def}}{=} R(a) \times S(x)$$

  for each $(a, x) \in A \times X$.

00Q9 **Proposition 9.3.6.1.2.** Let A, B, X, and Y be sets.

00QA 1. *Interaction With Converses.* Let

$$R\colon A \nrightarrow A,$$
$$S\colon X \nrightarrow X$$

   We have

$$(R \times S)^{\dagger} = R^{\dagger} \times S^{\dagger}.$$

00QB 2. *Interaction With Composition.* Let

$$R_1\colon A \nrightarrow B,$$
$$S_1\colon B \nrightarrow C,$$
$$R_2\colon X \nrightarrow Y,$$
$$S_2\colon Y \nrightarrow Z$$

   be relations. We have

$$(S_1 \diamond R_1) \times (S_2 \diamond R_2) = (S_1 \times S_2) \diamond (R_1 \times R_2).$$

[9] *Further Terminology:* Also called the **binary product of R and S**, for emphasis.

[10] That is, $R \times S$ is the relation given by declaring $(a, x) \sim_{R \times S} (b, y)$ iff $a \sim_R b$ and $x \sim_S y$.

Proof. *Item 1*, *Interaction With Converses*: Unwinding the definitions, we see that:

- We have $(a, x) \sim_{(R \times S)^\dagger} (b, y)$ iff:
  - We have $(b, y) \sim_{R \times S} (a, x)$, i.e. iff:
    - We have $b \sim_R a$;
    - We have $y \sim_S x$;
- We have $(a, x) \sim_{R^\dagger \times S^\dagger} (b, y)$ iff:
  - We have $a \sim_{R^\dagger} b$ and $x \sim_{S^\dagger} y$, i.e. iff:
    - We have $b \sim_R a$;
    - We have $y \sim_S x$.

These two conditions agree, and thus the two resulting relations on $A \times X$ are equal.
Item 2, *Interaction With Composition*: Unwinding the definitions, we see that:

- We have $(a, x) \sim_{(S_1 \diamond R_1) \times (S_2 \diamond R_2)} (c, z)$ iff:
  - We have $a \sim_{S_1 \diamond R_1} c$ and $x \sim_{S_2 \diamond R_2} z$, i.e. iff:
    - There exists some $b \in B$ such that $a \sim_{R_1} b$ and $b \sim_{S_1} c$;
    - There exists some $y \in Y$ such that $x \sim_{R_2} y$ and $y \sim_{S_2} z$;
- We have $(a, x) \sim_{(S_1 \times S_2) \diamond (R_1 \times R_2)} (c, z)$ iff:
  - There exists some $(b, y) \in B \times Y$ such that $(a, x) \sim_{R_1 \times R_2} (b, y)$ and $(b, y) \sim_{S_1 \times S_2} (c, z)$, i.e. such that:
    - We have $a \sim_{R_1} b$ and $x \sim_{R_2} y$;
    - We have $b \sim_{S_1} c$ and $y \sim_{S_2} z$.

These two conditions agree, and thus the two resulting relations from $A \times X$ to $C \times Z$ are equal. □

00QC 9.3.7 Products of Families of Relations

Let $\{A_i\}_{i \in I}$ and $\{B_i\}_{i \in I}$ be families of sets, and let $\{R_i \colon A_i \nrightarrow B_i\}_{i \in I}$ be a family of relations.

00QD **Definition 9.3.7.1.1.** The **product of the family** $\{R_i\}_{i \in I}$ is the relation $\prod_{i \in I} R_i$ from $\prod_{i \in I} A_i$ to $\prod_{i \in I} B_i$ defined as follows:

- Viewing relations as subsets, we define $\prod_{i\in I} R_i$ as its product as a family of sets, i.e. we have

$$\prod_{i\in I} R_i \overset{\text{def}}{=} \left\{ (a_i, b_i)_{i\in I} \in \prod_{i\in I} (A_i \times B_i) \;\middle|\; \begin{array}{l} \text{for each } i \in I, \\ \text{we have } a_i \sim_{R_i} b_i \end{array} \right\}.$$

- Viewing relations as functions to powersets, we define

$$\left[\prod_{i\in I} R_i\right]((a_i)_{i\in I}) \overset{\text{def}}{=} \prod_{i\in I} R_i(a_i)$$

for each $(a_i)_{i\in I} \in \prod_{i\in I} R_i$.

00R2 9.3.8 The Collage of a Relation

Let A and B be sets and let $R\colon A \nrightarrow B$ be a relation from A to B.

00R3 **Definition 9.3.8.1.1.** The **collage of** R[11] is the poset $\mathbf{Coll}(R) \overset{\text{def}}{=} \left(\mathrm{Coll}(R), \preceq_{\mathbf{Coll}(R)}\right)$ consisting of:

- *The Underlying Set.* The set $\mathrm{Coll}(R)$ defined by

$$\mathrm{Coll}(R) \overset{\text{def}}{=} A \coprod B.$$

- *The Partial Order.* The partial order

$$\preceq_{\mathbf{Coll}(R)}\colon \mathrm{Coll}(R) \times \mathrm{Coll}(R) \to \{\text{true}, \text{false}\}$$

on $\mathrm{Coll}(R)$ defined by

$$\preceq(a, b) \overset{\text{def}}{=} \begin{cases} \text{true} & \text{if } a = b \text{ or } a \sim_R b, \\ \text{false} & \text{otherwise.} \end{cases}$$

02AZ **Notation 9.3.8.1.2.** We write $\mathrm{Pos}_{/\Delta^1}(A, B)$ for the category defined as the pullback

$$\mathrm{Pos}_{/\Delta^1}(A, B) \overset{\text{def}}{=} \mathrm{pt} \underset{[A],\mathrm{Pos},\mathrm{ev}_0}{\times} \mathrm{Pos}_{/\Delta^1} \underset{\mathrm{ev}_1,\mathrm{Pos},[B]}{\times} \mathrm{pt},$$

[11] *Further Terminology:* Also called the **cograph of** R.

as in the diagram

$$\begin{tikzcd}
& \mathbf{Pos}_{/\Delta^1}(A,B) \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & \\
\mathbf{Pos}_{/\Delta^1} \underset{\mathbf{Pos}}{\times} \mathrm{pt} \arrow[dl] \arrow[dr] \arrow[dr, phantom, "\lrcorner"] & & \mathrm{pt} \underset{\mathbf{Pos}}{\times} \mathbf{Pos}_{/\Delta^1} \arrow[dl] \arrow[dr] \arrow[dl, phantom, "\lrcorner"] \\
\mathrm{pt} \arrow[dr, "{[A]}"'] & \mathbf{Pos}_{/\Delta^1} \arrow[dl, "\mathrm{ev}_{[0]}"] \arrow[dr, "\mathrm{ev}_{[1]}"'] & \mathrm{pt}. \arrow[dl, "{[B]}"] \\
& \mathbf{Pos} \quad\quad \mathbf{Pos} &
\end{tikzcd}$$

02B0 **Remark 9.3.8.1.3.** In detail, $\mathrm{Pos}_{/\Delta^1}(A,B)$ is the category where:

- *Objects.* An object of $\mathrm{Pos}_{/\Delta^1}(A,B)$ is a pair (X,ϕ_X) consisting of
  - A poset X;
  - A morphism $\phi_X\colon X \to \Delta^1$;

  such that we have

$$\begin{aligned}
\phi_X^{-1}(0) &= A,\\
\phi_X^{-1}(1) &= B.
\end{aligned}$$

- *Morphisms.* A morphism of $\mathrm{Pos}_{/\Delta^1}(A,B)$ from (X,ϕ_X) to (Y,ϕ_Y) is a morphism of posets $f\colon X \to Y$ making the diagram

$$\begin{tikzcd}
X \arrow[rr, "f"] \arrow[dr, "\phi_X"'] & & Y \arrow[dl, "\phi_Y"] \\
& \Delta^1 &
\end{tikzcd}$$

  commute.

00R4 **Proposition 9.3.8.1.4.** Let A and B be sets and let $R\colon A \nrightarrow B$ be a relation from A to B.

00R5 1. *Functoriality.* The assignment $R \mapsto \mathbf{Coll}(R)$ defines a functor

$$\mathbf{Coll}\colon \mathbf{Rel}(A,B) \to \mathrm{Pos}_{/\Delta^1}(A,B),$$

   where

- *Action on Objects.* For each $R \in \mathrm{Obj}(\mathbf{Rel}(A, B))$, we have

$$[\mathbf{Coll}](R) \overset{\text{def}}{=} (\mathbf{Coll}(R), \phi_R)$$

for each $R \in \mathbf{Rel}(A, B)$, where

  - The poset $\mathbf{Coll}(R)$ is the collage of R of Definition 9.3.8.1.1.
  - The morphism $\phi_R \colon \mathbf{Coll}(R) \to \Delta^1$ is given by

$$\phi_R(x) \overset{\text{def}}{=} \begin{cases} 0 & \text{if } x \in A, \\ 1 & \text{if } x \in B \end{cases}$$

for each $x \in \mathbf{Coll}(R)$.

- *Action on Morphisms.* For each $R, S \in \mathrm{Obj}(\mathbf{Rel}(A, B))$, the action on Hom-sets

$$\mathbf{Coll}_{R,S} \colon \mathrm{Hom}_{\mathbf{Rel}(A,B)}(R, S) \to \mathrm{Pos}(\mathbf{Coll}(R), \mathbf{Coll}(S))$$

of $\mathbf{Coll}$ at (R, S) is given by sending an inclusion

$$\iota \colon R \subset S$$

to the morphism

$$\mathbf{Coll}(\iota) \colon \mathbf{Coll}(R) \to \mathbf{Coll}(S)$$

of posets over Δ^1 defined by

$$[\mathbf{Coll}(\iota)](x) \overset{\text{def}}{=} x$$

for each $x \in \mathbf{Coll}(R)$.[12]

00R6 2. *Equivalence.* The functor of Item 1 is an equivalence of categories.

Proof. *Item 1*, *Functoriality*: Clear.
Item 2, *Equivalence*: Omitted. □

Appendices

[12] Note that this is indeed a morphism of posets: if $x \preceq_{\mathbf{Coll}(R)} y$, then $x = y$ or $x \sim_R y$, so we

A Other Chapters

Preliminaries

1. Introduction
2. A Guide to the Literature

Sets

3. Sets
4. Constructions With Sets
5. Monoidal Structures on the Category of Sets
6. Pointed Sets
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Relations

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Categories

11. Categories

Monoidal Categories

12. Constructions With Monoidal Categories

Bicategories

13. Types of Morphisms in Bicategories

Extra Part

14. Notes

References

[MO 460656] Emily. *Existence and characterisations of left Kan extensions and liftings in the bicategory of relations I*. MathOverflow. URL: https://mathoverflow.net/q/460656 (cit. on pp. 2, 3).

[MO 461592] Emily. *Existence and characterisations of left Kan extensions and liftings in the bicategory of relations II*. MathOverflow. URL: https://mathoverflow.net/q/461592 (cit. on pp. 3, 4).

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have either $x = y$ or $x \sim_S y$ (as $R \subset S$), and thus $x \preceq_{\mathbf{Coll}(S)} y$.